

MLDM Assignment 8

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 8

This are your test results

 Knowledge Questions

 Linear regression

Status

Seen but not answered

Your score

0 / 6

0%

Response

Select 'Right/Wrong' for each statement.

Note:

- Correct answer: 1 point
- Incorrect answer: -0.5 point
- Unanswered: 0 points
- Max score: 6 points
- Min score: 0 points

Unanswered	Right	Wrong	
☒	☐	☐	Multiple linear regression has more than one independent variable.
☒	☐	☐	Accuracy can be used as a metric to evaluate the performance of a linear regression model.
☒	☐	☐	The obtained solution is not unique.
☒	☐	☐	The learning rate has an influence on the rate of convergence of the gradient descent algorithm.
☒	☐	☐	By optimizing the cost function, We can find the parameters that gives the minimum difference between the actual and the predicted value.
☒	☐	☐	The outliers have no impact on the slope of the regression line.

☰ Perceptron

Status Not started

Your score 0 / 6 0%

Response

Select 'Right/Wrong' for each statement.

Note:

- Correct answer: 1 point
- Incorrect answer: -0.5 point
- Unanswered: 0 points
- Max score: 6 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	The linear decision boundary of the perceptron learning algorithm always passes through the origin.
☐	☐	☐	A perceptron has an input x_i with weight w_1 and a bias w_0 . $h(x)$ is the activation function used by the perceptron. The output of the perceptron is given by $h(w_1x_1 - w_0)$.
☐	☐	☐	XOR is a non-linear function.
☐	☐	☐	Kernel perceptron can be used on non-linear data.
☐	☐	☐	For perceptron classifiers with linearly separable data, the gradient descent algorithm may not converge for some initialization of weights.
☐	☐	☐	In a multilayer perceptron, the output of a layer will be the input of the next layer.

☰ SVM

Status Not started

Your score 0 / 6 0%

Response

Select 'Right/Wrong' for each statement.

Note:

- Correct answer: 1 point
- Incorrect answer: -0.5 point
- Unanswered: 0 points
- Max score: 6 points
- Min score: 0 points

Unanswered	Right	Wrong	
☐	☐	☐	SVM can be used for classification and regression.
☐	☐	☐	The optimal hyperplane in SVM has the minimum margin between the two classes.
☐	☐	☐	In SVM, a $(p+1)$ dimensional hyperplane is required to separate p dimensional data.
☐	☐	☐	For linearly separable data, the optimal hyperplane of SVM is unique.
☐	☐	☐	Kernel trick is a function to project the data into a low-dimensional space.
☐	☐	☐	For a multi-class classification problem, SVM can be applied by considering it as a multiple binary classification problem.

 Essay

Status

Not started

Your score

0 / 12

0%

Response

Explain the influence of the learning rate on the convergence of the gradient descent algorithm.

0 word

 Practice

 Perceptron

Status

Not started

Your score

0 / 15

0%

Response

A perceptron is trained on a small dataset. At an iteration t , the weight vector of the hyperplane $[w_1, w_2, w_0]^T = [2, 2, -4]^T$. The classifier classifies all the vectors correctly except two vectors $[1.8, 2.1]^T$ and $[0.1, 0.05]^T$. Your task is to calculate the weights of the vectors at $t+1$ iteration with a learning rate of 0.3. Provide your answer with two digits after the decimal points in the box given below.

$w_0 =$

$w_1 =$

$w_2 =$

 SVM

Status

Not started

Your score

0 / 10

0%

Response

Let x and y be two vectors. The two vectors are transformed into a high-dimensional space using a kernel function defined below. Compute $K(x, y)$ in order to fill the values for a and b.

$$\phi(z) = (1, \sqrt{2}z_1, \sqrt{2}z_2, z_1^2, z_2^2, \sqrt{2}z_1z_2)$$
$$K(x, y) = \langle \phi(x)^T, \phi(y) \rangle$$
$$K(x, y) = (a + (x \cdot y))^b$$

a=

b=

https://olat.vcrp.de/auth/RepositoryEntry/3909322134/CourseNode/106528667413088/test/0

3/4

Linear regression

Status

Not started

Your score

0 / 15

0%

Response

Calculate the regression coefficients of the best fit line ($y = w_0 + w_1 * x$) using the least square method for the data in the below table. Provide your answers up to two digits after the decimal point in the given box.

x	y
26	154
78	654
52	432
45	245
32	320

$w_0 =$

$w_1 =$

Programming

Upload file

Status

Not started

Your score

0 / 30

0%

Response

Download the files train.csv and test.csv and write the code to estimate the linear regression coefficients of the training data. Use the model to predict the test data and evaluate the model performance by using RMSE(root mean square as the metric).

Please upload your code here. Points will not be given if you do not upload your code and also make sure to print the output in the notebook cell while uploading.

As a preprocessing step, delete the rows that contain NaN values.

Note:

You are not allowed to use the **scikit-learn** library.

File

No file uploaded